

P4.py

```
import pandas as pd
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import punctuation
from nltk.stem import PorterStemmer
from sklearn.feature_extraction.text import CountVectorizer

data = pd.read_csv("sld2.csv")
print(data)

sw = stopwords.words("english")
ps = PorterStemmer()
def clean_review(txt):
    txt = txt.lower()
    txt = word_tokenize(txt)
    txt = [t for t in txt if t not in punctuation]
    txt = [t for t in txt if t not in sw]
    txt = [ps.stem(t) for t in txt]
    txt = " ".join(txt)
    return txt

data["clean_review"] = data["review"].apply(clean_review)
print(data)

cv = CountVectorizer()
vector = cv.fit_transform(data["clean_review"])
print(vector)

features = pd.DataFrame(vector.toarray(), columns = cv.get_feature_names_out())
print(features)
```

p5.py

```
import pandas as pd
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import punctuation
from nltk.stem import LancasterStemmer
from sklearn.feature_extraction.text import CountVectorizer

data = pd.read_csv("sld2.csv")
print(data)

sw = stopwords.words("english")
ls = LancasterStemmer()
def clean_review(txt):
    txt = txt.lower()
    txt = word_tokenize(txt)
    txt = [t for t in txt if t not in punctuation]
    txt = [t for t in txt if t not in sw]
    txt = [ls.stem(t) for t in txt]
    txt = " ".join(txt)
    return txt

data["clean_review"] = data["review"].apply(clean_review)
print(data)

cv = CountVectorizer()
vector = cv.fit_transform(data["clean_review"])
print(vector)

features = pd.DataFrame(vector.toarray(), columns = cv.get_feature_names_out())
print(features)
```

p6.py

```
import pandas as pd
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import punctuation
from nltk.stem import SnowballStemmer
from sklearn.feature_extraction.text import CountVectorizer
```

```
data = pd.read_csv("sld2.csv")
print(data)
```

```
sw = stopwords.words("english")
ss = SnowballStemmer("english")
def clean_review(txt):
    txt = txt.lower()
    txt = word_tokenize(txt)
    txt = [t for t in txt if t not in punctuation]
    txt = [t for t in txt if t not in sw]
    txt = [ss.stem(t) for t in txt]
    txt = " ".join(txt)
    return txt
```

```
data["clean_review"] = data["review"].apply(clean_review)
print(data)
```

```
cv = CountVectorizer()
vector = cv.fit_transform(data["clean_review"])
print(vector)
```

```
features = pd.DataFrame(vector.toarray(), columns = cv.get_feature_names_out())
print(features)
```

p7.py

```
import pandas as pd
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import punctuation
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import CountVectorizer

data = pd.read_csv("sld2.csv")
print(data)

sw = stopwords.words("english")
wnl = WordNetLemmatizer()
def clean_review(txt):
    txt = txt.lower()
    txt = word_tokenize(txt)
    txt = [t for t in txt if t not in punctuation]
    txt = [t for t in txt if t not in sw]
    txt = [wnl.lemmatize(t) for t in txt]
    txt = " ".join(txt)
    return txt

data["clean_review"] = data["review"].apply(clean_review)
print(data)

cv = CountVectorizer()
vector = cv.fit_transform(data["clean_review"])
print(vector)

features = pd.DataFrame(vector.toarray(), columns = cv.get_feature_names_out())
print(features)
```

NLP with Kamal Sir - 8369084928